

Kokkai NDL Pipeline Walkthrough

Content Extract — Immigration Stance Classification
from Diet Speeches (2012–2024)

datascience-publicpolicy-2026

10,961

speeches collected

9

immigration keywords

12 yrs

2012 – 2024

351 MB

raw text

KEYWORDS

技能実習 · 外国人労働者 · 出入国管理 · 特定技能

外国人材 · 不法滞在 · 多文化共生 · 移民政策 · 外国人犯罪

01 / THE PROBLEM

The Kokkai NDL search API returns rich per-speech metadata — but no stance, no canonical party split, and no electoral district. Our pipeline has to add those before the data is usable for analysis.



9 Immigration Keywords

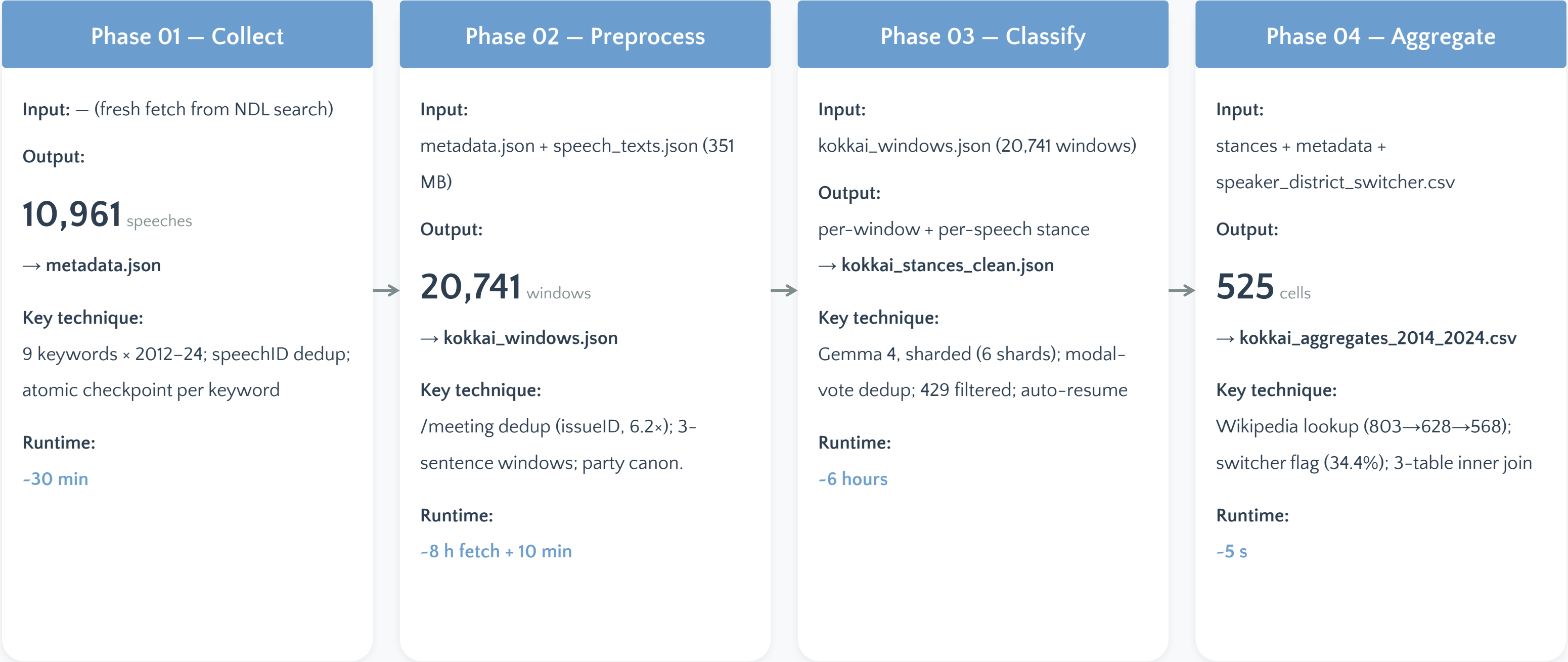
- 技能実習
- 外国人労働者
- 出入国管理
- 特定技能
- 外国人材
- 不法滞在
- 多文化共生
- 移民政策
- 外国人犯罪

Sample Metadata Record

speechID: 121614889X00320241219_027
speaker: 上月良祐
speakerYomi: こうづき りょうすけ
nameOfHouse: 参議院
nameOfMeeting: 内閣委員会
speakerGroup: 自由民主党
date: 2024-12-19
issueID: 121614889X00320241219
speechURL: kokkai.ndl.go.jp/txt/.../27

02 / PIPELINE OVERVIEW

Four phases, one join chain. Each phase reads the previous phase's output, transforms it in-memory, and writes a new on-disk artifact. The cross-team deliverable (`kokkai_aggregates_2014_2024.csv`) is produced at the end of phase 4.



03 / METADATA & EDA

Phase 1 paginates the NDL search API, dedupes by speechID, and atomically checkpoints per keyword. The resulting metadata.json is the join key for every later phase.

src/fetch_kokkai_metadata.py

```
KEYWORDS = ["技能実習","外国人労働者","出入国管理",  
            "特定技能","外国人材","不法滞在",  
            "多文化共生","移民政策","外国人犯罪"]  
FROM, UNTIL = "2012-01-01", "2024-12-31"  
  
# speechID-level dedup, accumulate keywords_matched
```

Top 5 Keywords

Keyword	Count
技能実習	6,141
特定技能	2,141
外国人労働者	1,829
外国人材	1,775
出入国管理	1,552

Top 5 Parties

Party	Count
自由民主党	1,763
自民・無所属	860
日本共産党	661
公明党	655
日本維新の会	432

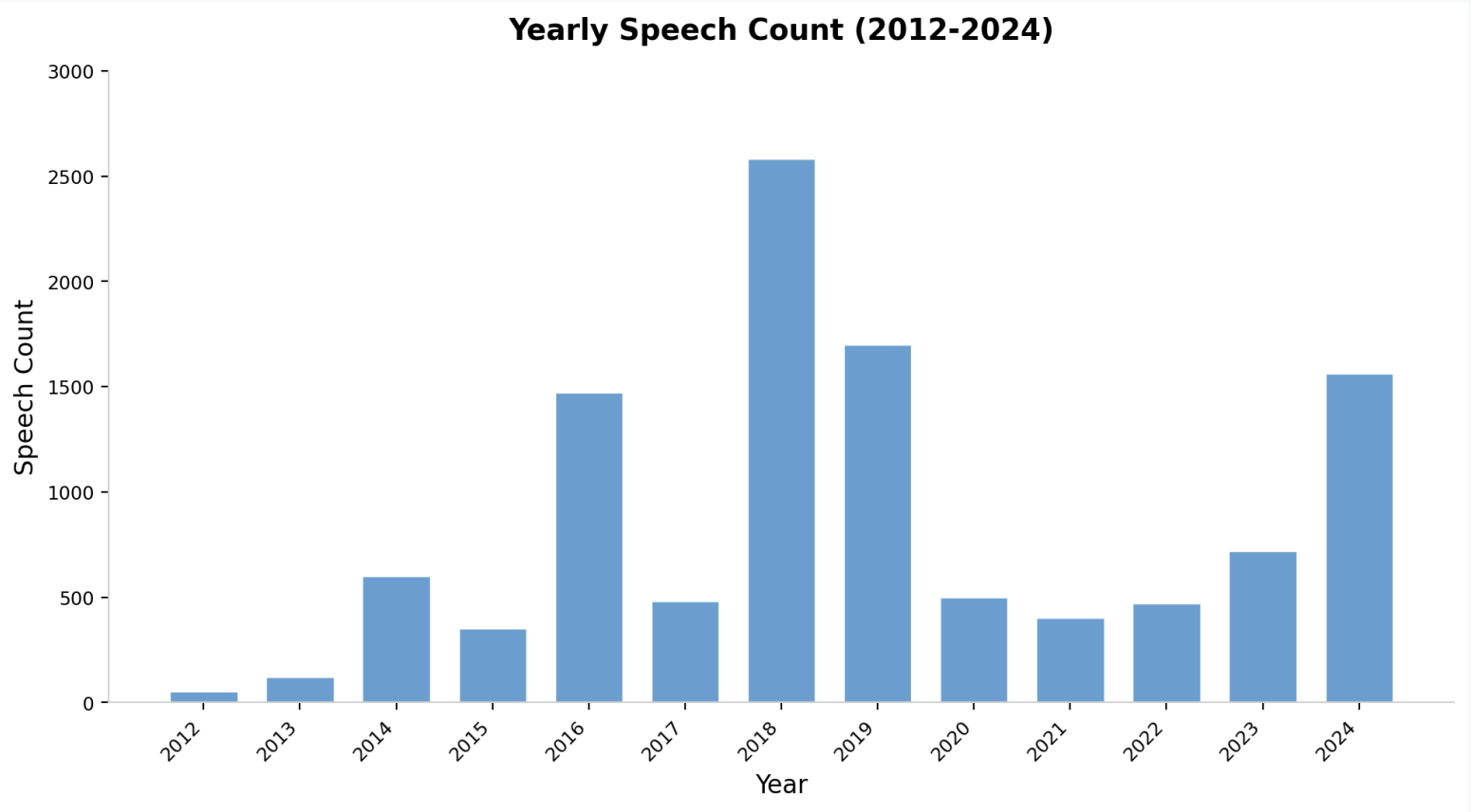
Field Normalization

NDL API Response: speechID, speech, nameOfHouse, nameOfMeeting, date, speaker, speakerGroup

metadata.json: speechID, speaker, speakerGroup, date, issueID, keywords_matched

Transformations: split issueID, drop null fields, aggregate keywords

Yearly Speech Count (2012–2024)



04 / TEXT & WINDOWING

Phase 2: Deduplicated transcript retrieval and keyword-centered context extraction.

src/fetch_kokkai_text.py — /meeting API dedup by issueID

```
# Strip _{speechOrder} → issueID
def derive_issue_id(speech_id):
    return speech_id.rsplit("_", 1)[0]

# 10,961 → 1,770 unique issueIDs (6.2× fewer)
def target_issue_ids(speeches, mode):
    party_ids = set()
```

src/clean_kokkai_text.py — raw text → 3-sentence windows

```
# Raw speech → split on 。 → find trigger → ±1 sentence
Raw speech → split on "。" → find trigger
→ extract sentence[n-1] + sentence[n] + sentence[n+1]
→ 3-sentence window with metadata
```

Sample Window (shortest)

「今だって続いているんですよ。これが技能実習生の実態ですよ。これ法務省の資料ですよ。」

/meeting API calls: 45,330 raw → 7,244 dedup'd

10,736 speeches with ≥1 window (avg 1.93 windows/speech)

Example — Window Output

```
{
  "windowID": "...056_0",
  "speechID": "...056",
  "triggers": ["出入国管理"],
  "sent_range": [n-1, n, n+1],
  "sentence_count": 3,
  "char_count": 237,
  "text": "...出入国管理法を改正する..."
}
```

Source: 西田譲 · 法務委員会 · 2013-05-10 · trigger: 出入国管理

Stats

20,741

windows generated

1.93

windows / speech

6.2×

fewer API calls

05 / CLASSIFY (Gemma 4)

Phase 3 calls Gemma 4 per window with a stance prompt, shards by md5(speechID) % 6, and aggregates per speech via modal vote.

Prompt Template — src/classify_stance_gemma.py

```
You are a deterministic, single-word classification API.  
Task: Classify this Japanese Diet speech on immigration  
into pro / neutral / anti. Output ONLY one line:  
  
{"stance": "pro"} OR  
{"stance": "neutral"} OR  
{"stance": "anti"}
```

Example — Input → Output

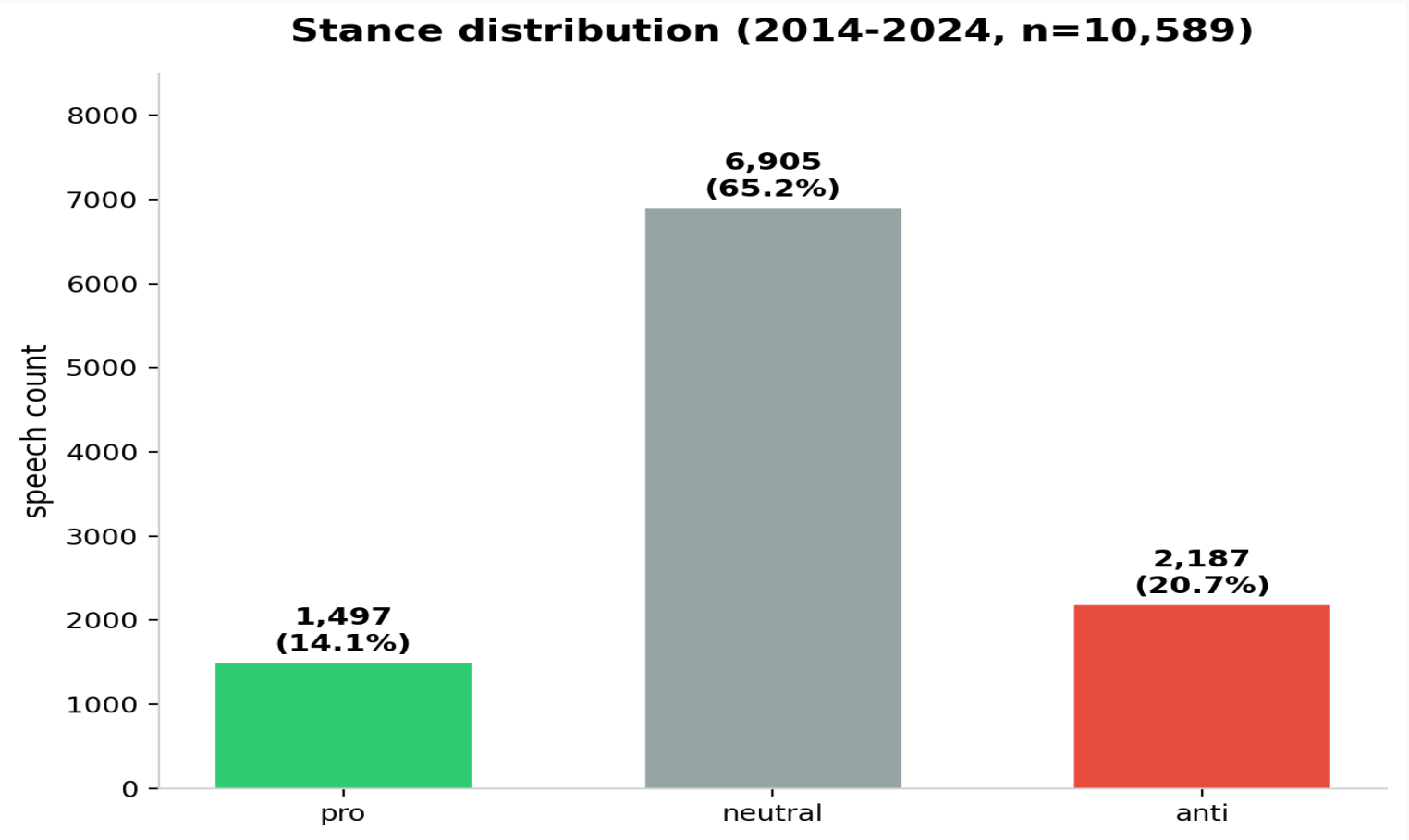
```
# Window (trigger: 多文化共生)  
「大臣は、所信の中で、多文化共生社会の  
実現を掲げられました。大変重要だと思います。」  
  
# Output  
{"stance": "pro"} ← Gemma 4-31b-it
```

Source: 藤野保史・法務委員会・2021-03-17

Top Triggers (20,741 windows)

Trigger	Count	Share
技能実習	11,311	54.5%
特定技能	3,248	15.7%
外国人労働者	2,730	13.2%

Stance Distribution (10,736 speeches)



pro: 1,523 (14.2%)

neutral: 7,001 (65.2%)

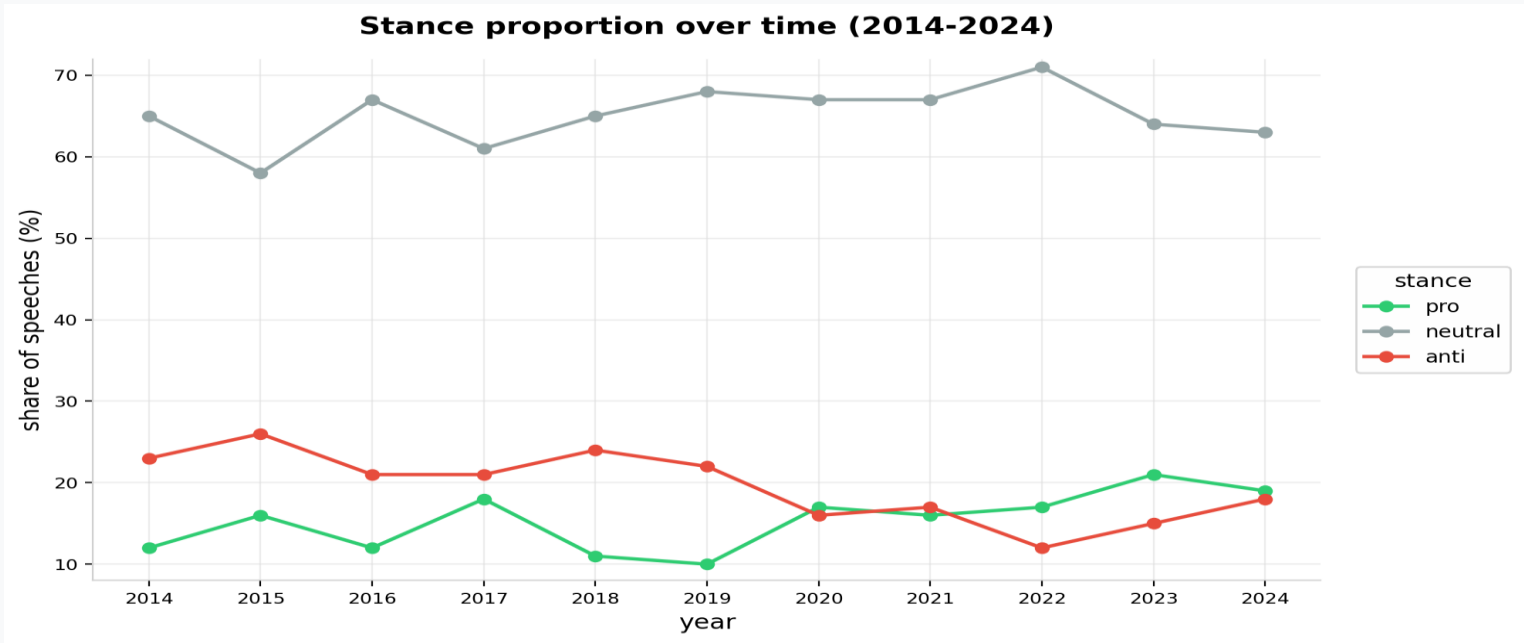
anti: 2,212 (20.6%)

anti 21% > pro 14% · neutral dominates · 2,065 windows have ≥2 triggers

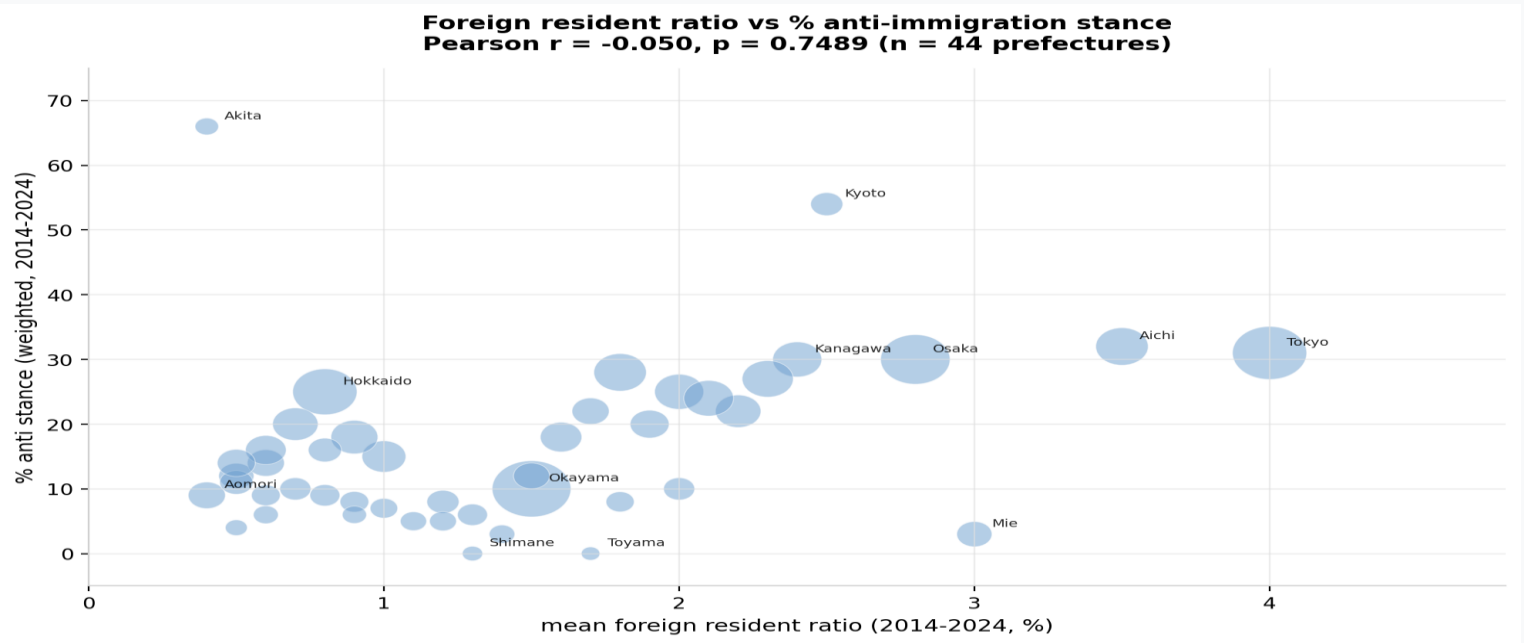
06 / ANALYZE

Spearman $\rho = -0.029$, $p = 0.85$ across 44 prefectures. Anti-stance share does **NOT** correlate with foreign-resident ratio.

Stance Proportion Over Time (2014–2024)



Foreign Resident Ratio vs % Anti Stance



Research Question Results

RQ: % anti stance vs foreign-resident ratio (44 prefectures)

Spearman $\rho = -0.029$, $p = 0.85$, $n = 44$

Not significant — no correlation found

Method: modal-vote per speech · FAILED_429 filtered · ties → neutral

Final CSV — kokkai_aggregates_2014_2024.csv

523 rows = (45 県 + 全国) × 11 years × 2 houses

Columns: year, prefecture, house, n, pro%, anti%, neut%, score, sw%

Sample:

2015, 岡山県, 衆議院, 16, 18.8, 43.8, 37.5, -0.25, 0.0

2015, 鳥取県, 参議院, 15, 53.3, 6.7, 40.0, +0.467, 0.0

Key insight: Anti share stays 18–23% across the decade. 2024 spike: 14% → 23%. No significant correlation between foreign-resident ratio and anti-immigration stance at prefecture level.

07 / LESSONS

Insight: Immigration stance ≠ policy stance — Our 1D label collapses two distinct dimensions.

Case Study — 仁比聡平 (JCP), #1 most-anti speaker (117/206)

2014-06-10 · 参議院 法務委員会 · 高度専門職ポイント制

「...新たな在留資格、高度専門職一、二を創設するものですが...年収要件が三百万円に引き下げられ...」

LLM → anti | He meant: Pro a *real* high-skilled visa, not this sham.

2018-11-28 · 参議院 本会議 · 特定技能 新在留資格

「...低賃金の人手不足分野への出稼ぎ労働と人材ビジネスの抜け道...四割を超えるに至っています。」

LLM → anti | He meant: Pro a deliberated visa protecting 128万 workers.

2024-06-14 · 参議院 本会議 · 永住者 在留資格取消事由の拡大

「...当事者は呼ばれず...エピソードとしてこんなひどいことがありましたよというだけのエビデンスで...」

LLM → anti | He meant: Pro more permanent-residency protections.

Takeaway: We lose the distinction between "I want fewer immigrants" and "I want a better visa for them." A 2D framework (policy critique × immigration support) would capture this nuance.

The 1D vs 2D Problem

1D label **anti** collapses two cells:
(anti-policy, pro-immig) U (anti-policy, anti-immig)

	pro-immig	neutral	anti-immig
pro-policy	pro	neutral	anti
neutral	pro	neutral	anti
anti-policy	仁比聡平	anti	anti

仁比聡平 falls in (anti-policy, pro-immig) — criticizing policy while supporting immigration.